

epipolar rectification, dense matching, and texture mapping. For dense matching, the authors adopted an existing network that was retrained using a synthetic dataset to learn spherical disparity and designed a spherical binoc-tree structure for depth map fusion. The proposed solution achieves superior performance when compared with traditional methods, as presented in Fig. 19(c). In conclusion, dense matching methods based on CNN networks attract attention in recent years. This review would not cover all aspects in this field.

III. APPLICATIONS

This section presents promising applications related to spherical images. Due to the characteristics of full FOV and low cost, spherical images have been used widely in a variety of applications. According to the purpose of this study, the reviewed applications would be restricted to 3D reconstruction, including cultural heritage documentation, urban modeling and navigation, tunnel mapping and inspection, and other applications, e.g., urban tree localization, emergency response and rescue, and underwater occlusion avoidance.

A. Cultural heritage documentation

Cultural heritage documentation (CHD) is the earliest usage of spherical images, which can be dated back to two centuries ago [18]. In contrast to the large-scale acquisition required in aerial photogrammetry, the scale of cultural heritage documentation is much smaller but with serious occlusions, e.g., building pillars and inner structures. It requires close-range data acquisition with a large FOV to decrease human labor and ensure completeness. When compared to conventional aerial plane-based and recent UAV-based photogrammetry, spherical images can be recorded in a more flexible way, e.g., ground-fixed tripods and hand-held poles, and have been widely used for cultural heritage documentation.

In the field of photogrammetry and remote sensing, [15], [134], [135] have made an earlier contribution to CHD by using spherical images. This work presents the basic principle of the proposed spherical photogrammetry (SP), i.e., the collinear equation of spherical imaging and the coplanarity condition for image orientation. By using two church architectures, this study verifies the advantages of spherical images with the remarkable efficiency of data acquisition, the completeness of documentation, and the low economic cost compared with other photogrammetric instruments.

With the explosive development of consumer-grade spherical cameras, recent researchers have also turned to adopting low-cost cameras for data acquisition and 3D reconstruction of CHD. For example, [9] used a Panono 360 camera for the image collection of two churches, which is a ball-shaped camera that consists of 36 camera submodules to record surrounding environments. This study verified the centimeter-level precision of dense point clouds from SfM and MVS-based photogrammetric processing. [136] exploited a Xiaomi Mi Sphere 360 camera for the precision assessment of 3D modeling. In this study, a Leica TS30 total station has been used for an in-site survey of ground control points (GCPs), and two software packages, including Agisoft Metashape and

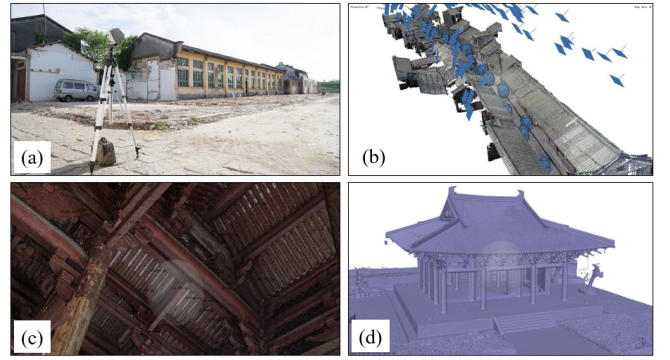


Fig. 20. Cultural heritage documentation by using aerial UAV images and terrestrial spherical images [22]. (a) Data acquisition based on the Weiss AG Civetta camera; (b) image orientation of aerial-ground images; (c) dense matching of building inner structures; (d) 3D reconstruction of buildings.

Pix4dMapper, have been evaluated. Experiments demonstrate that millimeter-level accuracy has been obtained in both image orientation and 3D reconstruction by using GCPs and laser scanning point clouds.

Because of the complex structure of cultural heritages, aerial and ground images have also been combined to reconstruct complex cultural heritages. [25] combined low-altitude UAV images and hand-held spherical images for 3D documentation of a bell tower. In this work, a DJI Phantom 4 Pro UAV has been adopted for scanning external structures under a properly designed flight trajectory, and two spherical cameras, i.e., the GoPro Fusion and the Kandao Qoocam 8K, have been used for scanning internal structures. Based on an SfM and MVS-based processing solution, the reconstructed 3D model has high-quality details in the external and internal of the complex bell tower. In contrast to using consumer-grade spherical cameras, [22] conducted high-quality digital twin documentation of a well-known town, Long Hu Gu Zhai at Guangdong Province, China. This study uses professional Weiss AG Civetta spherical cameras for terrestrial images with 230 megapixels resolution and a UAV for along-street aerial images. By using the Agisoft Metashape, 3D models were reconstructed from aerial and terrestrial images, as presented in Fig. 20. The results demonstrate the potential of aerial and ground images for the precision documentation of complex buildings.

B. Urban modeling and navigation

Spherical cameras have been integrated with mobile mapping systems (MMS) for urban street modeling and navigation. The most famous application of spherical images comes from Google Street View, which provides immersive navigation and observation along urban streets. At the beginning of Google Street View, collected street images have also been used for 3D modeling of building facades in addition to image navigation [24], [8], as shown in Fig. 21(a). [137] designed a unified framework to reconstruct 3D models from spherical images captured from a Ladybug camera, in which bundle adjustment free image orientation, piecewise planarity constrained dense matching, and novel depth map fusion were reported consid-

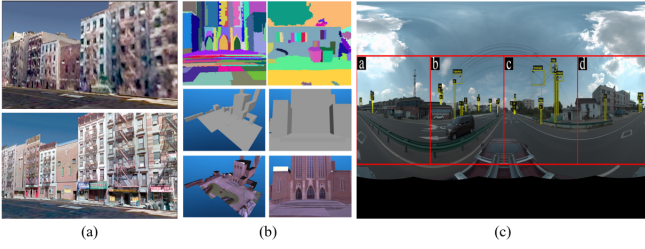


Fig. 21. Urban modeling and navigation. (a) mesh models from Google Street View [24]; (b) 3D models from spherical images [132]; (c) street object localization [140].

ering the characteristics of low-texture, repetitive pattern, and large light changes in urban streets.

In contrast to mesh-based 3D models, recent studies show well-formed wire-frame models reconstructed from spherical images. Under the Manhattan world assumption, [132] converted spherical images into cubic-map representation and segmented urban scenes into planar structures under the constraints of image color, and edge and normal information from MVS-based depth maps, as presented in Fig. 21(b). By using street images and deep learning techniques, [138] designed a workflow for building detection and height calculation and created 3D models for large-scale urban scenes.

Due to the high image overlap of large FOV cameras, spherical images have also been widely used to achieve localization and navigation in complex urban environments. [13] proposed a solution for the large-scale localization of photos without geotagged labels. In this study, 3D sparse models reconstructed from street view images were used as the reference data source, and a three-step algorithm was designed for geo-localization, which includes image retrieval-based coarse geo-localization, reliable feature matching between the query image and retrieved candidate images, and the PnP (Perspective n Points) based precise geo-localization. Instead of using low-level features, [139] turned to detect high-level semantic information from spherical images, such as lamp posts and street signs, which are detected based on a retrained YOLO CNN network. To geo-localizing interesting targets within urban scenes to assist vehicle navigation, [140] proposed a line of bearing (LOB) based positioning method for urban street objects, e.g., road lamps, shown in Fig. 21(c).

C. Tunnel mapping and inspection

Artificial tunnels are extensively used in daily life, e.g., traffic tunnels and drainage pipes. Regular inspection is essential to ensure normal functions and extend the lifespans. In practice, the inspection work has been widely achieved through labor-sensitive operators. In recent years, with the advent of small-size spherical cameras and the development of image processing techniques, image-based inspection solutions have also been presented and verified to achieve automatic tunnel mapping and inspection [141].

In the work of [142], a low-cost and flexible system has been designed to create a panorama image of the traffic tunnels, in which SfM-based image orientation and stitching-based image registration techniques have been used under the constraint

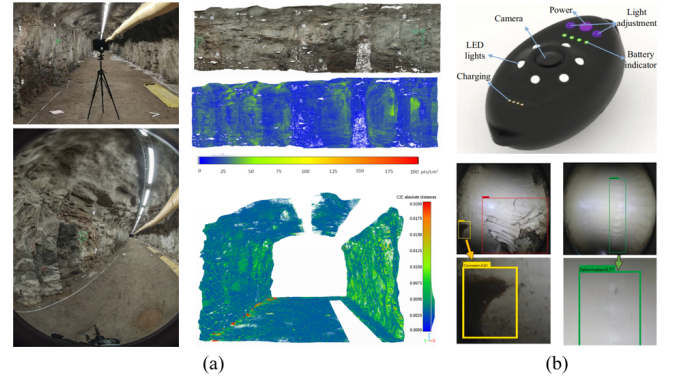


Fig. 22. Tunnel mapping and inspection. (a) mine tunnel mapping [143]; (b) drainage pipe inspection [146].

of tunnel design data. This work verified the feasibility of image-based mapping for tunnel in-field inspection. Based on the rapid data acquisition ability of spherical cameras, [143] designed an overall framework for photogrammetric tunnel mapping. In this work, an Insta360 Pro sphere camera integrating six fisheye camera modules has been used to record images with 4000 by 3000 pixels. During data acquisition, the spherical camera was installed on a tripod to image recording with long exposure times. For quality verification, point clouds were also collected by using a Riegl VZ-400i TLS scanner. Experimental results demonstrate that the reconstructed tunnel models can achieve millimeter level precision when compared with TLS point clouds, as presented in Fig. 22(a).

In contrast to large-size tunnel mapping, small-size gas and drainage pipes are the other typical inspection scenes that require the full FOV acquisition ability of a spherical camera due to the limited space in these pipes. In the work of [144], drainage pipe inspection has been implemented by using low-cost GoPro Fusion video cameras. Key frames were extracted from recorded videos and converted to cubic-map images, then used to reconstruct 3D models based on typical SfM and MVS-based solutions. For the inspection of drainage pipes with minimal space, [145] designed a pipeline capsule machine (PCM), as presented in Fig. 22(b), which mainly consists of a large FOV camera, a LED (light-emitting diode) lighting instrument, and a power supply module. The PCM sensor can record images with the flow of water, which are further used for crack and erosion detection based on deep learning-based methods [146].

Similarly, for the inspection of gas pipes, [147] designed an inspection robot that is equipped with a catadioptric camera for spherical image collection. Unlike the PCM, depending on flowing water, the inspection robot can operate on other pipes since it uses wheels for forward moving. In a word, spherical cameras have good advantages for tunnel mapping and inspection due to the low cost of sensors and the limited space of tunnels.

D. Other applications

In recent years, spherical images have also been used in other applications, e.g., emergency response and rescue, urban

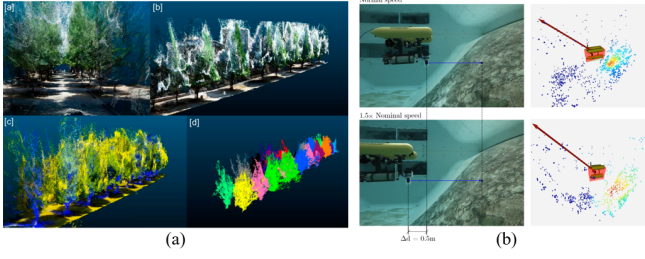


Fig. 23. Spherical cameras for (a) urban tree detection [148] and (b) obstacle avoidance of underwater vehicles [150].

tree detection and mapping, collision avoidance of underwater vehicles, etc. [11] designed an integrated photogrammetric system for rapid data acquisition and damaged building inspection. This work took advantage of low-altitude UAVs and ground backpack MMSs to collect aerial-ground images and validate the integration of aerial-ground images for the full 3D reconstruction of urban buildings to boost the observation completeness.

Tree number, location, and structure are critical for the precision management of vegetation resources. Although conventional aerial photogrammetry and LiDAR (Light Detection and Ranging) can provide large-scale RS data, they suffer serious occlusions of urban scenes and cannot achieve accurate measurements of street trees. Considering these issues, [148] proposed using spherical cameras for the 3D modeling and parameter calculation of urban trees. In this study, 3D models were reconstructed based on SfM, which were further processed to extract individual trees, as shown in Fig. 23(a). Based on the 3D models, some useful parameters, such as trunk diameter and tree height, can be then calculated. Similarly, [149] adopted a deep learning network for individual tree detection from urban street images and depended on the single-view depth estimation and triangulation techniques to calculate the precise tree geo-localization.

The underwater image collection is a frequent task for marine exploration, which has been completed by using autonomous underwater vehicles. Acoustic sensors are widely utilized instruments for obstacle avoidance. However, they cannot work well at a close range. In the work of [150], an obstacle detection instrument has been designed by using a spherical camera, which can generate real-time point clouds from a visual SLAM system and assist the autonomous obstacle detection, as presented in Fig. 23(b).

IV. PROSPECTS

3D reconstruction from spherical images has been greatly promoted by the development of professional and consumer-grade cameras and automatic image processing techniques, as well as their increasing usage in varying fields. Compared with classical perspective images, there still exist some challenges for 3D reconstruction based on spherical images, which can be categorized as exploitation of crowdsourcing image, spatial resolution of spherical image, camera distortion and calibration, and integration of aerial-ground image. The details are presented in the following subsections.



Fig. 24. Crowdsourcing images from Google Street Views.

A. Exploitation of crowdsourcing image

In recent years, spherical images can be easily recorded based on low-cost and easy-to-use spherical cameras, as presented in Fig. 4. Besides, spherical images can also be downloaded from well-known map providers, e.g., Google Maps and Tencent Maps. These crowdsourcing images are usually freely provided for non-commercial usage, which plays a critical role in spherical image-based applications [12]. However, 3D reconstruction based on these crowdsourcing images is non-trivial for two main reasons. On the one hand, the diversity of crowdsourcing images is very large, which can be recorded by using professional or consumer-grade cameras, and obtained from indoor or outdoor scenes, as shown in Fig. 24; on the other hand, the overlap degree of crowdsourcing images can not be ensured since they are captured by non-professional users or just for the visual navigation purpose. These factors can frequently cause the failure of 3D reconstruction.

Crowdsourcing images still have great potential in 3D reconstruction, especially for urban buildings, although they have the above-mentioned drawbacks. There are some possible solutions to exploit the potential of freely available crowdsourcing images. First, image classification methods can be used to separate indoor and outdoor images, e.g., recent deep learning-based networks. The classified images can be used for subsequent indoor or outdoor 3D modeling. Second, existing crowdsourcing images can be useful compensation from the street viewpoints, such as for UAV images from aerial viewpoints. In this situation, crowdsourcing images could be registered to the reconstructed 3D models from UAV images, avoiding the requirement of high overlap degrees in image-based 3D reconstruction.

B. Spatial resolution of spherical image

Spherical cameras record the 360 by 180 degrees of surrounding environments using only one shot, which dramatically accelerates data acquisition. However, because of the limited CCD (Charge Coupled Device) dimensions of cameras, the spatial resolution of collected images is also obviously decreased when compared with classical perspective cameras with images of the same CCD dimensions. The spatial resolution causes a contradiction between the efficiency of data acquisition and the quality of the reconstructed model.

- **The efficiency of data acquisitions.** Much more time would be consumed to record images with higher resolutions to ensure the enough camera exposure time. For